

CUSTOMER RETENTION & CHURN ANALYSIS REPORT

Prepared By:

Sahinka Sarkar

Project Type:

Customer Retention & Churn Analytics Project

Tools Used:

Microsoft Excel | Python | Power BI

Dataset:

Telco Customer Churn Dataset

TABLE OF CONTENTS

1. Executive Summary	1
2. Business Problem Statement	1
3. Project Objectives	2
4. Dataset Overview	2
5. Tools & Technologies Used	3
6. Methodology	3
7. Key Performance Indicators (KPIs)	9
8. Churn Analysis	9
9. Retention Analysis	10
10. Cohort Analysis	10
11. Customer Lifetime Analysis	11
12. Dashboard Features	11
13. Business Insights	14
14. Business Recommendations	14
15. Challenges Faced	15
16. Learning Outcomes	15
17. Future Improvements	15
18. Conclusion.....	16

1. EXECUTIVE SUMMARY

This project focuses on analyzing customer churn and retention behavior for a subscription-based business using the Telco Customer Churn dataset. The analysis was performed using Excel, Python, and Power BI to transform raw customer subscription data into meaningful business insights and an interactive retention analytics dashboard.

The project evaluated customer demographics, contract types, internet services, payment methods, customer tenure, monthly charges, and churn behavior to identify key retention trends and business growth opportunities.

The analysis revealed that customers with month-to-month contracts, electronic check payment methods, and shorter tenure periods were more likely to churn. On the other hand, long-term contract customers and customers using automatic payment methods demonstrated stronger retention behavior.

The final Power BI dashboard provides a professional and interactive solution for monitoring customer retention trends, churn risks, customer lifetime patterns, and business decision-making.

2. BUSINESS PROBLEM STATEMENT

Subscription-based businesses face major challenges in retaining customers and reducing churn. High customer churn negatively impacts business revenue, increases customer acquisition costs, and reduces long-term profitability.

Without a proper analytics system, identifying high-risk customer segments, understanding churn behavior, and monitoring retention trends becomes difficult.

This project aims to build a data-driven analytical solution that helps business stakeholders:

- Identify churn reasons
- Understand retention behavior
- Monitor customer lifetime trends
- Improve long-term customer loyalty
- Support strategic customer retention decisions

3. PROJECT OBJECTIVES

The primary objectives of this project were:

- Analyze overall customer churn and retention performance
- Identify high-risk customer segments
- Evaluate customer retention trends
- Perform cohort-style retention analysis
- Analyze customer lifetime behavior
- Build an interactive Power BI dashboard
- Generate actionable business recommendations

4. DATASET OVERVIEW

The project uses the Telco Customer Churn dataset containing customer demographic information, subscription details, service usage patterns, payment methods, tenure information, and churn status.

Dataset Columns :

- Customer ID
- Gender
- Senior Citizen
- Partner
- Dependents
- Tenure
- Phone Service
- Multiplelines
- Internet Service
- Online Security
- Online Backup
- Device Protection
- Tech Support
- Streaming TV
- Streaming Movies
- Contract
- Paperless Billing
- Payment Method
- Monthly Charges
- Total Charges
- Churn

Dataset Information :

The dataset contains customer records across multiple subscription plans and service categories. It includes customer behavior attributes that help analyze retention patterns and churn risks.

5. TOOLS & TECHNOLOGIES USED

Tool	Purpose
Microsoft Excel	Data cleaning, validation, Pivot Tables
Python	Data analysis and visualization
Power BI	Interactive dashboard creation and visualization

6. METHODOLOGY

The project followed a structured analytics workflow.

Step 1 — Data Cleaning in Excel

The initial data cleaning and preprocessing process was performed using Microsoft Excel Power Query Editor.

The following tasks were completed:

- Removed duplicate records
- Handled missing values
- Corrected data types
- Verified numerical fields
- Standardized categorical values
- Built Pivot Tables for preliminary churn and retention analysis

All cleaned data and Pivot Tables were stored in separate sheets within the same Excel workbook for further processing and dashboard integration.

Pivot Tables Preview:

Contract Type	Average of Churn_Flag
Month-to-month	42.71%
One year	11.28%
Two year	2.85%
Grand Total	26.58%

Payment Method	Average of Churn_Flag
Bank transfer (automatic)	16.73%
Credit card (automatic)	15.25%
Electronic check	45.29%
Mailed check	19.20%
Grand Total	26.58%

Internet Services	Average of Churn_Flag
DSL	19.00%
Fiber optic	41.89%
No	7.43%
Grand Total	26.58%

Average Customer Lifetime
<u>32.42178612</u>

Step 2 — Data Processing and Analysis Using Python

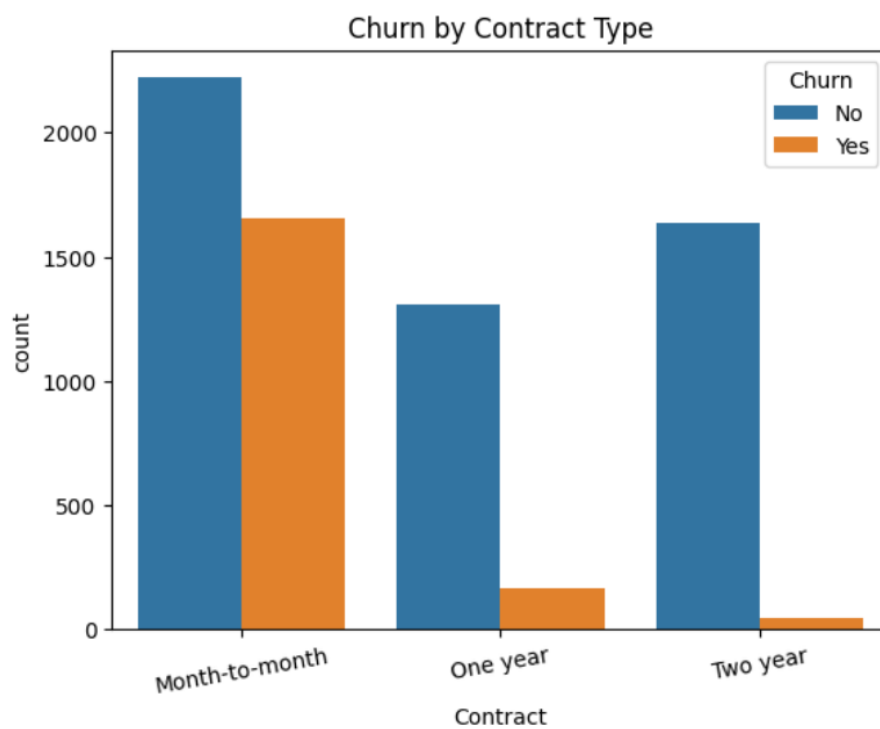
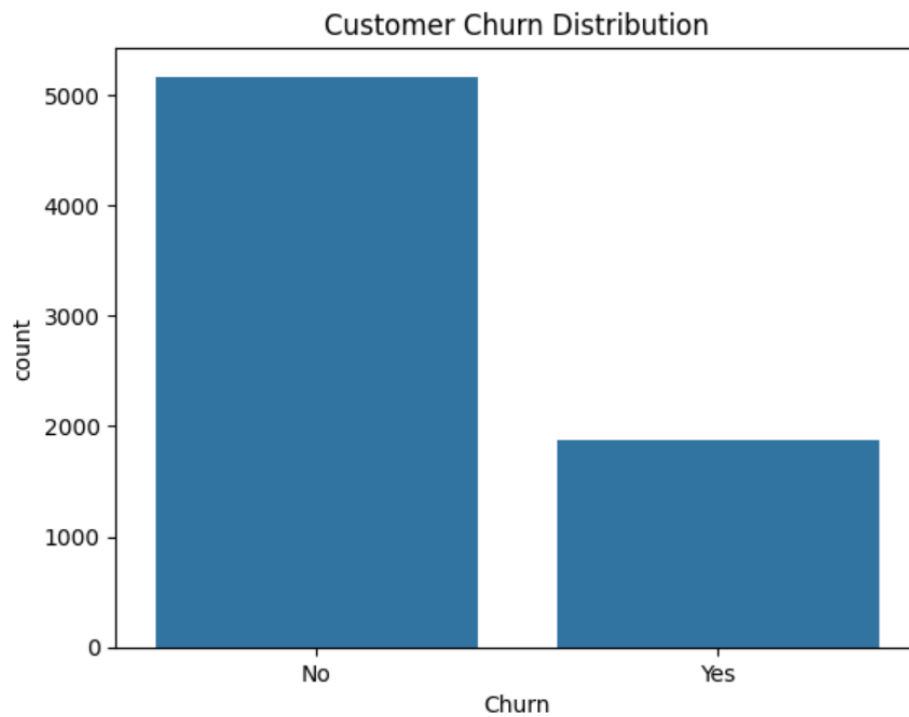
Python was used for advanced data processing, customer churn analysis, retention analysis, and visualization. Libraries such as pandas, matplotlib, and seaborn were used to perform analytical calculations and generate business insights.

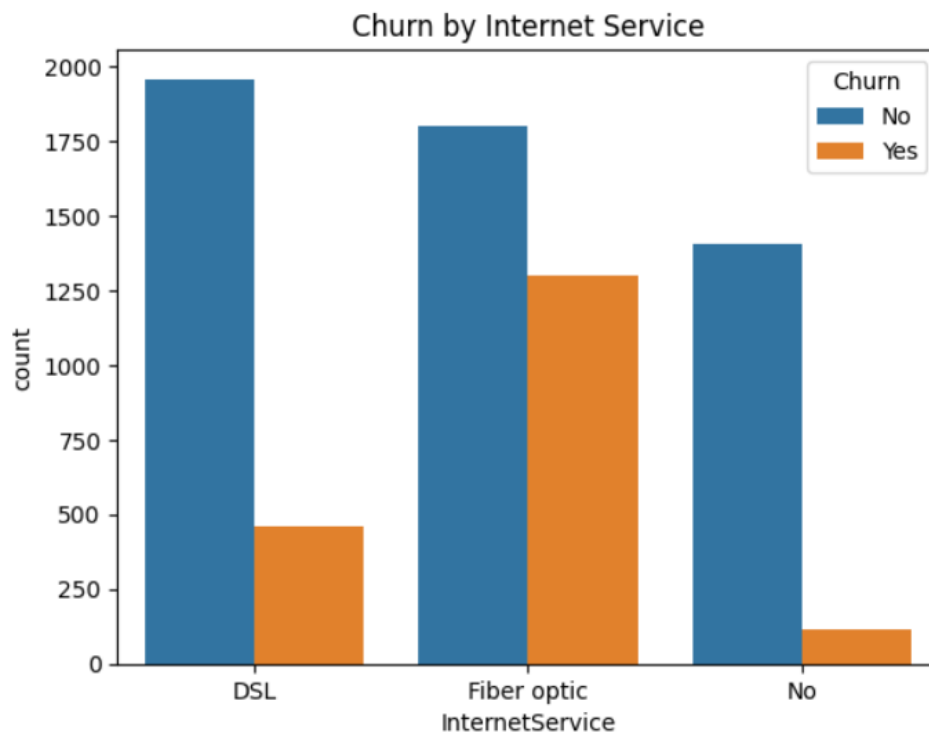
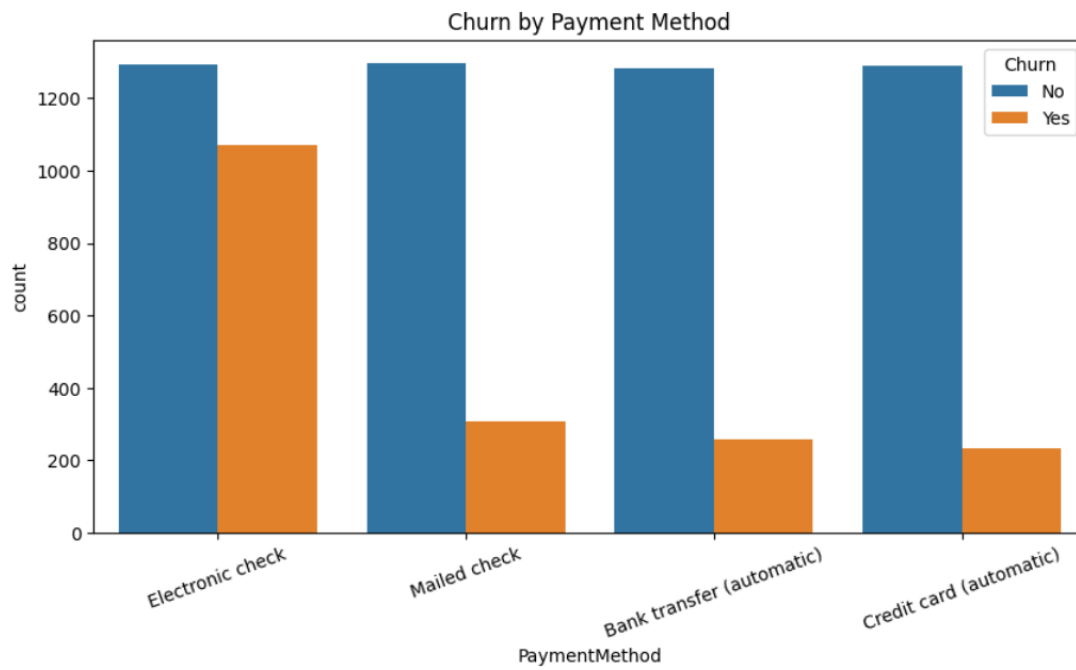
The following tasks were completed using Python:

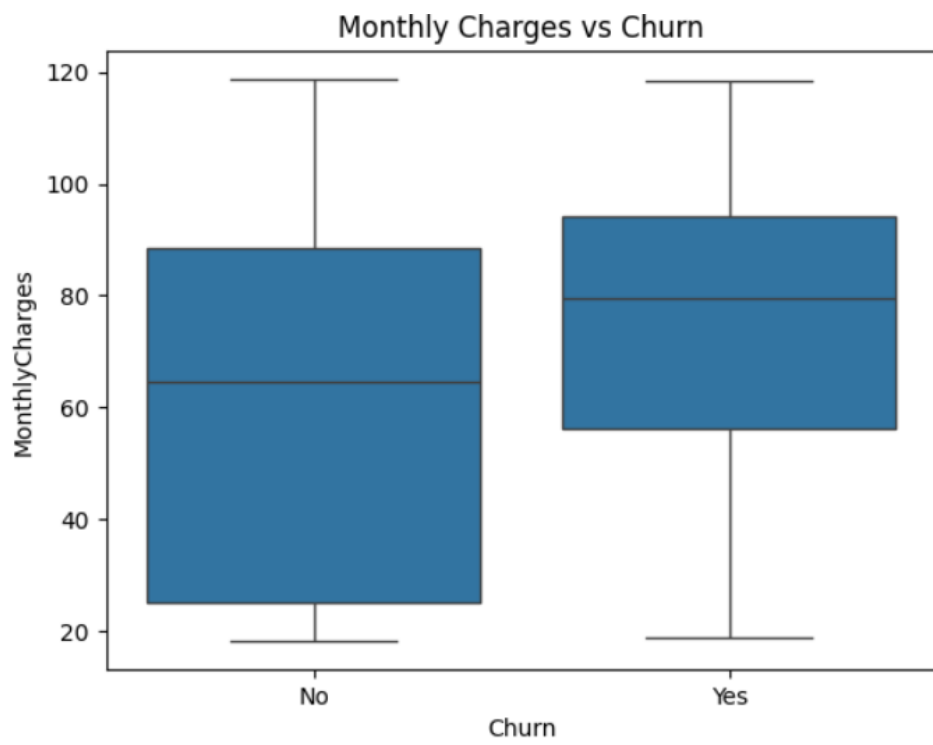
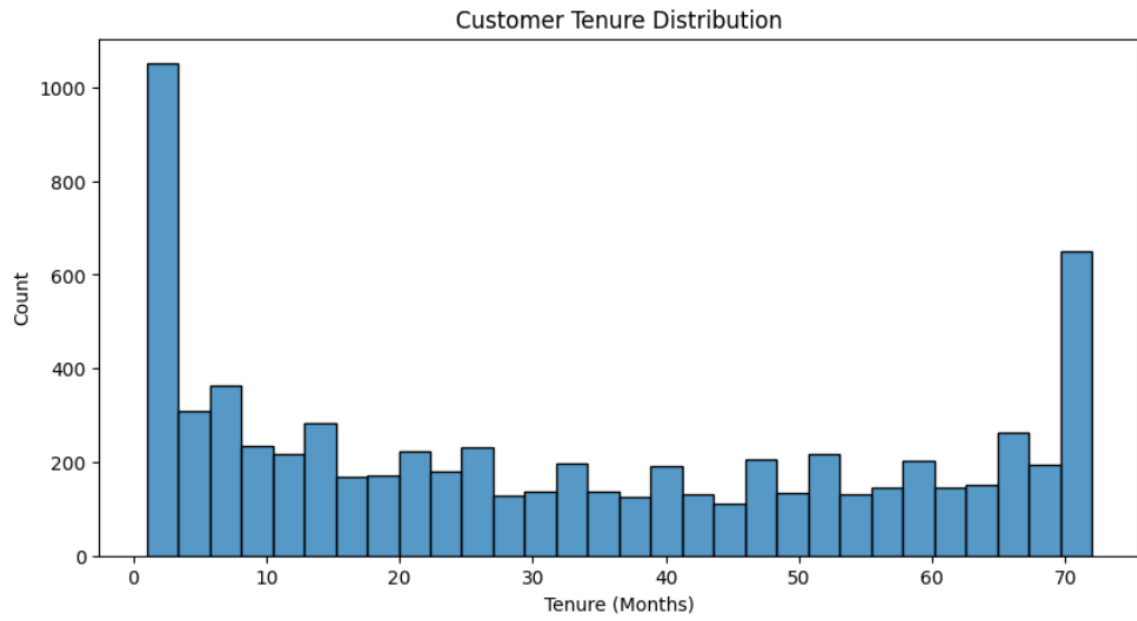
- Data preprocessing and transformation
- Churn rate analysis
- Retention calculations
- Cohort-style retention analysis
- Customer lifetime analysis
- Data visualization and trend analysis
- Exporting processed data for Power BI dashboard integration

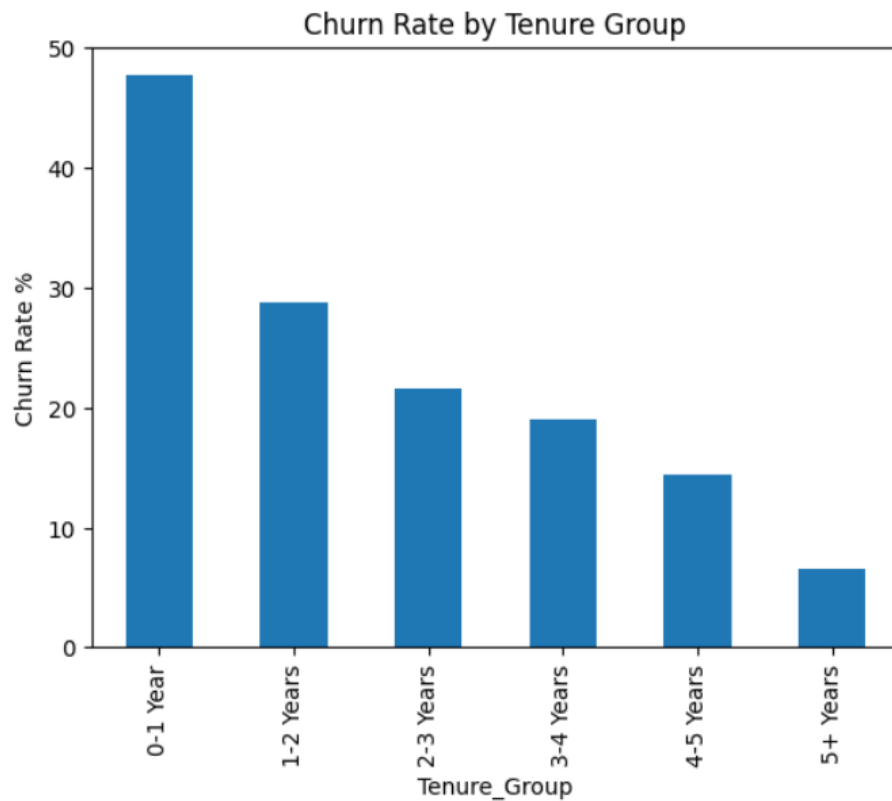
Python helped automate analytical calculations and enabled deeper exploration of customer behavior and retention trends.

Python Charts Preview:









Step 3 — Dashboard Development in Power BI

An interactive dashboard was built in Power BI to visualize:

- Churn KPIs
- Retention KPIs
- Customer lifetime analysis
- Cohort retention trends
- Contract and payment behavior
- Customer segmentation insights

The dashboard allows users to interactively explore customer retention behavior using filters and slicers.

7. KEY PERFORMANCE INDICATORS (KPIs)

The following KPIs were analyzed:

KPI	Description
Total Customers	Total customer count
Churned Customers	Total churned customers count
Retained Customers	Total retained customers count
Churn Rate	Percentage of customers lost
Retention Rate	Percentage of customers retained
Average Customer Lifetime	Average customer tenure
Average Monthly Charges	Average subscription charges

8. CHURN ANALYSIS

The business demonstrated significant customer churn among specific customer groups and subscription categories.

Churn by Contract Type

Customers with month-to-month contracts showed the highest churn rate compared to one-year and two-year contract customers.

Insight:

Long-term contracts improve customer loyalty and reduce churn risk.

Churn by Payment Method

Customers using electronic check payment methods showed significantly higher churn behavior.

Insight:

Automatic payment methods contribute to better customer retention and stability.

Churn by Internet Service

Fiber optic customers demonstrated higher churn rates compared to DSL users.

Insight:

Certain service categories may require quality improvements and better customer engagement.

Churn by Customer Tenure

Customers within the first year showed the highest churn probability, while long-term customers demonstrated stronger loyalty.

Business Interpretation:

Early-stage customer engagement plays a major role in customer retention success.

9. RETENTION ANALYSIS

The retention analysis focused on identifying customer loyalty trends and long-term customer engagement patterns.

Retention by Tenure Group

Customers with longer tenure demonstrated significantly higher retention rates.

Insight:

Retention improves steadily as customers remain active on the platform.

Retention by Contract Type

Two-year contract customers demonstrated the highest retention performance.

Business Interpretation:

Long-term subscription plans create stable customer relationships.

Retention by Payment Method

Customers using automatic payment methods showed better retention behavior compared to electronic check users.

Retention by Tech Support

Customers with tech support services demonstrated stronger retention and lower churn rates.

Insight:

Customer support services improve customer satisfaction and long-term engagement.

10. COHORT ANALYSIS

A cohort-style retention analysis was performed using tenure groups and contract categories to evaluate customer retention behavior over time.

The analysis showed:

- Retention improves steadily with longer customer tenure
- Long-term customers demonstrate stronger loyalty
- Customer engagement increases retention probability
- Short-tenure customers are the most vulnerable customer segment

Business Interpretation:

Customer retention strategies should focus heavily on the first year of customer engagement.

11. CUSTOMER LIFETIME ANALYSIS

Customer lifetime patterns were analyzed using customer tenure and monthly subscription charges.

The analysis revealed that customers who remain active longer generate significantly higher long-term business value.

Customer Lifetime Insights

- Average customer lifetime is approximately 32 months
- Long-term customers contribute higher revenue
- Customer loyalty improves over time
- Retention directly impacts customer lifetime value

Customer Lifetime Value Formula

$$CLV = \text{Monthly Charges} \times \text{Customer Tenure}$$

12. DASHBOARD FEATURES

An interactive two-page dashboard was developed using Power BI to visualize customer churn, retention trends, cohort analysis, and customer lifetime behavior.

The dashboard was designed with interactive visualizations, KPI cards, filters, and slicers to support business decision-making and improve user experience.

The two dashboard pages were connected using Power BI Page Navigation buttons, allowing smooth navigation between:

- Churn Overview Dashboard
- Customer Retention & Cohort Analysis Dashboard

This navigation feature improved dashboard usability and created a professional multi-page reporting experience similar to real-world business intelligence dashboards.

Page 1: Churn Overview Dashboard

The Power BI dashboard includes:

KPI Cards

- Total Customers
- Churn Rate
- Average Lifetime (Months)
- Average Monthly Charges

Visualizations

- Churn Rate by Contract Type – Stacked column Chart
- Churn Rate by Payment Method – Stacked Bar Chart
- Customers by Internet Service & Churn – Stacked Column Chart
- Customer Churn Distribution – Donut Chart
- Churn Rate by Tenure Group – Line Chart

Interactive Features

- Tech Support Slicer
- Tenure Group Slicer
- Device Protection Slicer
- Page Navigation Button

Page 2: Customer Retention & Cohort Analysis Dashboard

The Power BI dashboard includes:

KPI Cards

- Churned Customers
- Churn Rate
- Retention Rate
- Retained Customers

Visualizations

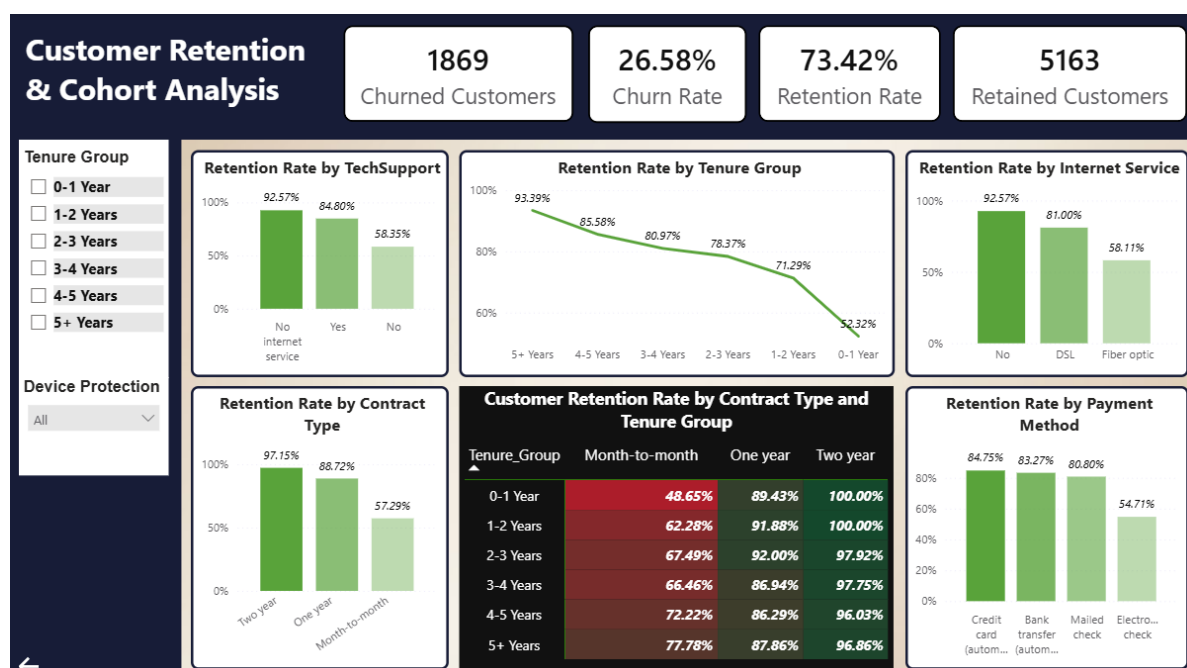
- Retention Rate by TechSupport – Stacked column Chart
- Retention Rate by Payment Method – Stacked Column Chart
- Retention Rate by Contract Type – Stacked Column Chart
- Retention Rate by Internet Service – Stacked Column Chart
- Customer Retention Rate by Contract Type and Tenure Group – Cohort Heatmap
- Retention Rate by Tenure Group – Line Chart

Interactive Features

- Tenure Group Slicer
- Device Protection Slicer
- Page Navigation Button

The dashboard enables business users to interactively monitor customer churn and retention performance.

Dashboard Preview



13. BUSINESS INSIGHTS

The following important business insights were identified:

- Month-to-month customers have the highest churn risk
- Electronic check users are more likely to churn
- Long-term contract customers demonstrate stronger retention
- Customers with longer tenure contribute higher business value
- Automatic payment methods improve retention performance
- Tech support services improve customer loyalty
- Customer churn is highest during early customer lifecycle stages

These insights can help businesses optimize customer retention strategies and improve long-term profitability.

14. BUSINESS RECOMMENDATIONS

1. Improve Customer Onboarding

Focus on improving the onboarding experience during the first year to reduce early customer churn.

2. Promote Long-Term Contracts

Provide discounts and incentives for yearly and two-year subscription plans.

3. Encourage Automatic Payments

Offer benefits for customers using automatic payment methods.

4. Improve Customer Support

Provide proactive support for high-risk customer groups.

5. Launch Customer Loyalty Programs

Reward long-term customers with retention offers and loyalty benefits.

6. Improve High-Churn Services

Investigate dissatisfaction among Fiber optic customers and improve service quality.

15. CHALLENGES FACED

During the project, several challenges were encountered:

- Handling missing and inconsistent data
- Understanding customer churn behavior
- Creating meaningful retention metrics
- Designing interactive dashboard layouts
- Managing multiple tools and workflows
- Performing cohort-style retention analysis

These challenges improved analytical thinking and problem-solving skills.

16. LEARNING OUTCOMES

Through this project, the following skills were developed:

- Data cleaning and preprocessing
- Customer analytics
- Churn and retention analysis
- Cohort analysis
- Power BI dashboard development
- Business storytelling
- Data visualization
- Professional report writing

This project significantly improved practical business analytics skills.

17. FUTURE IMPROVEMENTS

The project can be further enhanced by:

- Adding machine learning churn prediction models
- Performing advanced customer segmentation
- Integrating real-time customer databases
- Developing automated ETL pipelines
- Adding predictive retention analytics
- Including customer satisfaction analysis

These enhancements would provide deeper business intelligence capabilities.

18. CONCLUSION

This project successfully transformed raw customer subscription data into meaningful business insights using Excel, Python, and Power BI.

The analysis revealed important patterns in customer churn, retention behavior, customer lifetime trends, and cohort performance. Long-term contracts and automatic payment methods emerged as strong retention drivers, while short-tenure customers and electronic check users demonstrated higher churn risk.

The final Power BI dashboard provides a professional and interactive analytical solution that supports data-driven customer retention strategies and business decision-making.

This project demonstrates strong practical skills in:

- Data Analytics
- Python
- Power BI
- Customer Analytics
- Business Intelligence
- Dashboard Design
- Reporting & Visualization

The project also reflects the ability to present technical analysis in a professional, business-focused format suitable for internships, portfolios, academic submissions, and business presentations.